

DIT047/DAT357 Requirements Engineering Final Exam

March 18, 2026

Examiner/Contact Person

Jennifer Horkoff

Jennifer will come to the rooms to check for questions at roughly 14:30, 16:00, and 17:00.

Alternatively Phone: 0733 050517 (call again if no answer, I am in another exam room)

Authorized Aids

Writing instruments (pens, pencils), English Dictionary. No textbook or notes.

Grading Scale for Exams

% Grade	Final Grade
0-49	Fail (U)
50-64	3
65-79	4
80-100	5

PLEASE OBSERVE THE FOLLOWING:

- All answers must be in English. Answers must be legible and readable.
- Sort the questions in order before handing them in.
- Put the number of the question on every page.
- This exam has 7 pages.

Part 1: Multiple item and Short-Answer Questions

Question 1: Multiple Choice Questions (15 points)

On your paper, write the name of the question, then the letter(s) of your chosen answer(s), e.g., 1.1: c., 1.3: a, d, e. For each question, there may be more than one correct answer. For full marks, list all the correct answers. Each correct answer listed is +1 point, each incorrect answer is -1 point.

1. Which of these requirements captures design information and is therefore **not** well-formed?
Note: these requirements focus on a bike rental system. In this question there may be more than one correct answer, list them all.
 - (a) The mobile app shall use a map interface built with Google Maps API to show bike locations.
 - (b) The system shall display bike availability within 5 seconds of a user request.
 - (c) The system shall record the start and end time of each rental.
 - (d) The system shall allow users to find the nearest available bike.
 - (e) The system shall use a three-tier architecture separating UI, business logic, and data storage.
 - (f) The system shall store all rental records in a MySQL relational database
2. Which of the following statements about creative ideas is correct? One or more may be correct.
 - (a) Creative ideas are defined as ideas that are novel and implementable.
 - (b) Creative ideas are defined as ideas that are simple and novel and useful.
 - (c) Creative ideas are defined as ideas that are well-known and useful.
 - (d) All creative ideas become requirements.
 - (e) Creative ideas may be implemented by one or more requirements.
 - (f) Creative ideas are best generated in groups.
 - (g) Creative ideas are best generated alone.
3. When conducting requirements elicitation as part of agile, what challenges commonly hold? One or more may be correct.
 - (a) Stakeholders are too technical.
 - (b) Stakeholders being too busy to participate regularly.
 - (c) Lack of working software.
 - (d) One stakeholder cannot easily represent all stakeholders.
 - (e) Stakeholders feel responsible for requirements.
 - (f) Requirements should not act as legal contracts.
4. Which of the following definitions about safety-related concepts is correct? One or more may be correct.
 - (a) Risk is the probability of occurrence of harm.
 - (b) Risk is the combination of the probability of occurrence of harm and the severity of that harm.
 - (c) Unreasonable risk is when risk is judged to be unacceptable in a certain context according to valid societal moral concepts.

- (d) Unreasonable risk is the combination of the probability of occurrence of harm and the severity of that harm.
 - (e) Safety is the absence of risk.
 - (f) Safety is the absence of unreasonable risk.
5. Which statements about security requirements concepts are true? One or more may be correct.
- (a) Security goals state desires.
 - (b) Security requirements state desires.
 - (c) Security goals have a narrow scope.
 - (d) Security requirements have a narrow scope.
 - (e) Security goals are not testable.
 - (f) Security requirements are not testable.

Question 2: Short-Answer Questions (20 points)

1. List two requirements elicitation techniques. List one advantage and disadvantage of each. (4 point)
2. These requirements are not verifiable (testable). For each requirements, why are they not verifiable? Rewrite each to be verifiable. (4 points)
 1. The system shall provide an easy-to-use interface for renting bikes.
 2. The bike rental system shall display bike information.
3. In the lectures, we covered several templates and structured natural language approaches for capturing requirements. Name two of them. Name one benefit and one drawback of using a template or structured natural language representation for capturing requirements. (4 points)
4. Explain, both backwards and forwards requirements traceability. (4 points)
5. In the guest lectures you saw various examples of how different types of AI could be used to improve activities in requirements engineering. Give two examples of how AI can be used to help requirements engineering activities. Be as specific as possible. Describe one drawback of each of these examples. (4 points)

Part 2: Domain Example and Long-Answer Questions

The remaining questions on the exam will relate to a problem in a domain, as described below. The scenario describes the situation today and the expectations for the new system. Focus your analysis and modeling on the envisioned (to be) system, but keep in mind the problems and requests with the as is situation, trying to avoid problems and satisfy user needs.

Campus Classroom Security

The Current Situation: At a university located next to a local high school, classes are often disrupted by high school students who gather outside the classrooms. These students make loud noise in the hallways, bang on the doors and windows, and would sometimes enter classrooms if the doors were not kept locked. This creates an unsafe and distracting environment for university students and faculty, making it difficult to hold lectures and maintain a calm learning atmosphere.

To address this problem, the university currently uses a security guard provided by a private security agency. However, this solution is expensive, and even with a guard present, the high school students still gather nearby and sometimes continue to make noise. When no guard is on site, it can take more than 45 minutes for one to arrive after being called, which leaves classes unprotected during disturbances. Another challenge is that the university has no easy way to keep track of what incidents occur over time. Reports to the university administration, the high school leadership, and the security provider are difficult to prepare because information is scattered, incomplete, or reported via email without a consistent record.

Vision (System To Be): The university plans to introduce a new digital system to improve the handling of security incidents and reduce classroom disruptions. If no security guard is present, users should be able to use the system to call one directly. In more serious situations, the system should also offer a fast way to contact the police. Because users may not always know the right response to a disturbance, the system should give clear guidance on what actions to take: when to stay inside the classroom, when to call a guard, and when a situation is serious enough to alert the police.

This system will allow university students, faculty, and security guards to record incidents as they occur, using either a mobile app or a web interface. Video and/or audio can be recorded or uploaded. Related incidents, such as disruptions caused by the same group of students recorded by many users, should be linked together by an administrator so patterns can be understood more clearly.

Over time, the system will store a full history of all incidents and generate statistics that can be used to create summaries and official reports for the university, for the nearby high school, and for the security agency. These reports can help all parties understand the level of disruption, evaluate trends, and decide on long-term actions.

Due to the safety-critical context, the system must be fast and readily available. The system must follow GDPR rules to protect personal data and must meet the Web Content Accessibility Guidelines so it is usable by all members of the university community. The goal is to have the system fully ready and available for use by January 2027.

Question 3: Context Diagram (10 points)

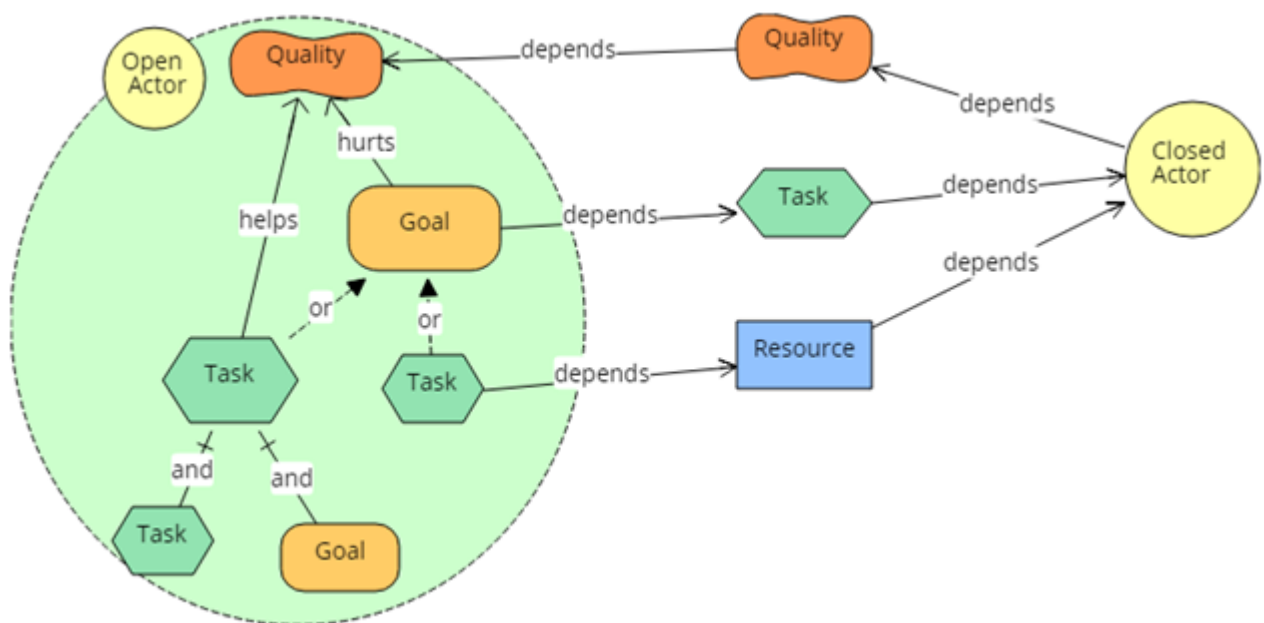
For the given case, draw a context diagram. Add the system actor, identify the relevant stakeholder/actor entities, and the relationships between the system and these entities, labelled with high-level inputs and outputs. Try to capture four actors (the system actor counts), and appropriate relationships. Supplement the diagram with text to explain any ambiguous or unclear parts of the model.

Question 4: Goal Model (10 points)

For the given case, draw a goal model. Identify the relevant actors, including the system actor. Try to capture at least four actors, and important dependencies between actors. Show the internal goals/tasks/resources/qualities for at least one actor, the others can be “closed” showing only dependencies to and from the actors. See the legend below for open and closed actors. For the open actor, capture the desired goals, tasks, resources, and qualities. Add some internal relationships between these elements.

There are many elements that can be included in the model. Aim for: four actors including one open actor with an actor boundary, at least one goal, quality, resource, task, four dependency links, two contribution links and two AND/OR links. In other words, the model does not have to be complete as per the domain description. It's helpful to use the information already captured in the other diagrams; however, we won't mark you on consistency between diagrams. Supplement the diagram with text to explain any ambiguous or unclear parts of the model.

Goal Model Legend (note, can use alternative shapes and labels as long as the mapping to the original concepts is clear)



Question 5: Customer Journey Map (10 points)

Draw a customer journey map for the given case. Indicate the persona that the journey applies to. Think about the channels and touchpoints that the persona would have to go through when using the system. Try to include at least three channels and five touchpoints.

Question 6: Textual Requirements (20 points)

Write SRS requirements and user stories for the case provided. List four functional requirements in SRS form, four non-functional requirements in SRS form, two domain assumptions, and two constraints. In addition, list four user stories, each with acceptance criteria. The user stories should be unique, i.e. not a repeat of one of the SRS requirements. Remember the desired characteristics of user stories and SRS requirements when writing your requirements.

Question 7: Prototype (10 points)

Draw two prototype screens for the given case implementation. The screens can include pop-up windows, just be clear how and why they appear. The purpose of the screens should be clear. The screen should implement at least two functional requirements (SRS or user stories). These requirements should be clearly implemented on the screens. Write the full text of the requirements implemented here, so we do not have to flip back to find them. For each screen, write: the requirements implemented, and explain how and exactly where the requirements are implemented.

Question 8: User Testing Tasks (5 points)

Come up with a list of five test tasks for the users to perform during usability tests for the designed screens. Each screen should be tested by at least two tasks. For these tasks, indicate which of your screens they test. The tasks should cover some of the main use cases or requirements of your system. The role of the user should be clear. Tasks should conform to the desirable qualities of user test tasks as discussed in the lecture.